



The Scots College

Year 12 Mathematics Extension 2

Pre-Trial

25th March 2011

Name: _____

General Instructions

- Working time - 2 hours
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Standard Integrals Table attached

TOTAL MARKS:

THERE ARE 4 QUESTIONS.

WEIGHTING: 30 %

Question 1: (Marks 20)

a)

Using De Moivre's Theorem, or otherwise, find the square root of $\sqrt{3} + i$.

2

b)

Sketch on separate Argand diagrams, the locus of z defined by:-

i) $|z - 2 - 3i| = 4$

2

ii) $\text{Arg} \left(\frac{z-i}{z+1} \right) = 0$

2

iii) $z\bar{z} = z + \bar{z}$

4

iv) $z + z^{-1}$ given that $z + z^{-1}$ is real.

4

c)

Given the equation $(5 + 3i)z^2 - (1 - 4i)z + 8 - 2i = 0$

Find:-

3

- i) The product of the roots.
- ii) The modulus of your answer in part i)
- iii) The argument of your answer in part i)

d)

If z is a point on the unit circle with $\arg(z) = \alpha$

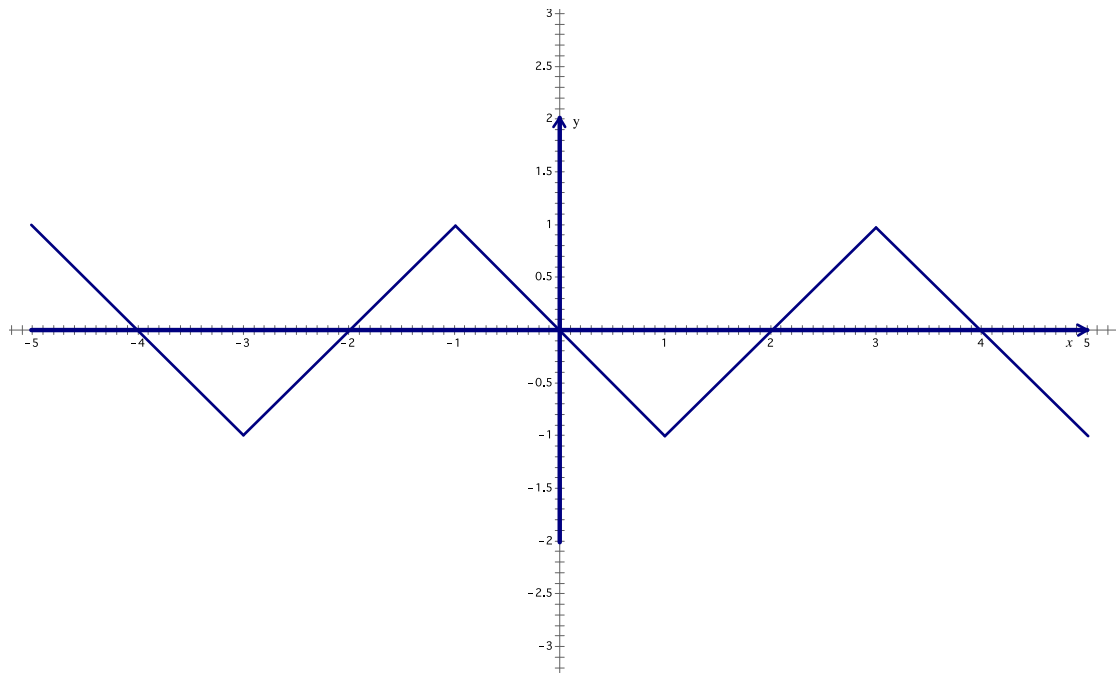
3

- i) Prove that $|z^2 - z| = |z - 1|$
- ii) Find the arguments of z^2 and $z^2 - z$ in terms of α .

Question 2: (23 Marks) Start a New Booklet

a)

The graph of $y = f(x)$ for $-5 \leq x \leq 5$ is shown below.



On separate axes, sketch the following curves. Indicate clearly any turning points, asymptotes and intercepts with the coordinate axes.

i) $y = f(|x|)$

2

ii) $y = \frac{1}{f(x)}$

2

iii) $y = \sqrt{f(x)}$

2

iv) $y = x f(x)$

2

b)

i) Sketch the graph of $y = \frac{9x - 27}{x^2 - x - 2}$, clearly indicating any asymptotes and any points where the graph intercepts the axes. 4

ii) Find the equation of the normal to the curve $x^3 + 6xy + y^3 = 1$ at $(1, 0)$ 3

c)

i) Show that the function $f(x) = \ln(\ln x)$ is an increasing function for all values of x in its domain. 2

ii) On separate diagrams, sketch the curves

$\alpha) y = f(x)$ 2

$\beta) y = \ln(|\ln x|)$ 2

$\gamma) y = |\ln|\ln x||$ 2

Question 3: (20 Marks) Start a New Booklet

a)

Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$

i) Find the coordinates of its foci S and S' . 2

ii) Find the equations of its directrices. 2

b)

i) Find the equation of the tangent to the ellipse $4x^2 + 9y^2 = 36$ at the point $(1, \frac{4\sqrt{2}}{3})$. 2

ii) Show that this tangent passes through the point $(6, \frac{1}{\sqrt{2}})$. 2

c)

i) If $y = mx + c$ is a tangent to the Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then show that $c^2 = a^2 m^2 - b^2$. 3

ii) The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Prove that $PS + PS' = 2a$ where, S and S' are the two foci of the ellipse. 3

d)

Find the equation of that diameter which bisects the chord $7x + y - 20 = 0$ of the hyperbola $\frac{x^2}{3} - \frac{y^2}{7} = 1$ 6

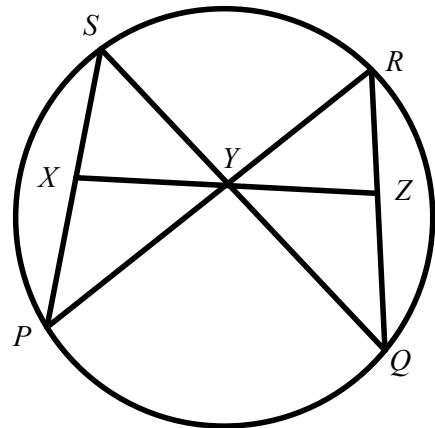
Question 4: (15 Marks)

a)

$PQRS$ is a cyclic quadrilateral. The diagonals PR and QS intersect at right angles at Y .

X is the midpoint of PS .

XY produced meets QR at Z .



i) Copy the diagram onto your answer scripts.

Show that $XY = SX$.

ii) Hence show that XZ perpendicular to QR .

b)

i) The equation $2x^3 - x^2 - 6x + 3$ has roots α , β and γ .

Evaluate $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$

2

ii) The roots of the equation $x^3 + 3x^2 + 7x + q = 0$ are in arithmetic progression.

Find the value of the constant q .

2

c)

Prove by mathematical induction $2^{2^n} \geq 5^{2^n}$ for $n \geq 5$.

4

END OF EXAM

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

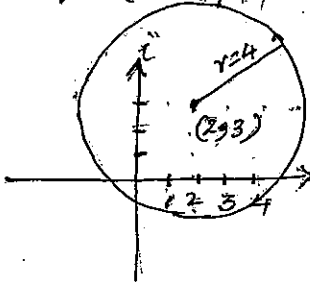
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

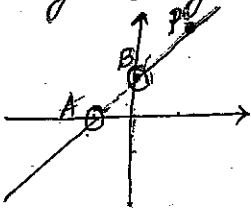
NOTE : $\ln x = \log_e x, \quad x > 0$

Q1 (a) Let $Z = \sqrt{3} + i$
 $= 2 \left(\frac{\sqrt{3}}{2} + \frac{i}{2} \right)$
 $= 2 \cos \frac{\pi}{6}$
 $Z^{1/2} = 2^{1/2} \cos \frac{\pi}{12}$
 $= \pm \sqrt{2} \cos \pi/12$

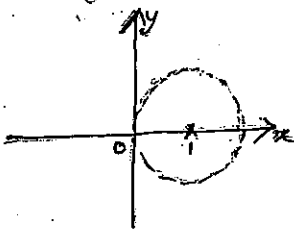
(b) (i) $|Z - 2 - 3i| = 4$
 $|Z - (2 + 3i)| = 4$



(ii) $\text{Arg} \left(\frac{Z-i}{Z+1} \right) = 0$
 $\text{Arg}(Z-i) = \text{Arg}(Z+1)$



(iii) $Z\bar{Z} = Z + \bar{Z}$
 $(x+iy)(x-iy) = x+iy+x-iy$
 $x^2+y^2 = 2x$
 $x^2-2x+1+y^2=1$
 $(x-1)^2+y^2=1$

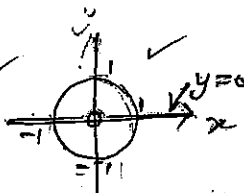


(iv) $Z + Z^{-1}$
 $(x+iy) + \frac{1}{x+iy}$
 $x+iy + \frac{1}{x+iy} \times \frac{x-iy}{x-iy}$
 $x+iy + \frac{x-iy}{x^2+y^2}$
 $\frac{x+x}{x^2+y^2} + i \left(\frac{y}{x^2+y^2} - \frac{y}{x^2+y^2} \right)$

$\therefore \text{Real} \therefore \frac{y}{x^2+y^2} = 0$

$y(x^2+y^2-1) = 0$

$y=0$ or $x^2+y^2=1$



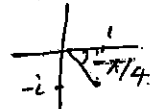
excluding 0

(c) $(5+3i)z^2 - (1-4i)z + 8-2i = 0$

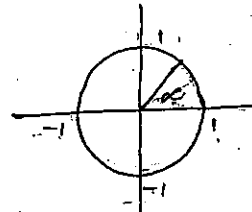
(i) Product of roots $= \frac{8-2i}{5+3i} \times \frac{5-3i}{5-3i}$
 $= \frac{40-24i-10i-6}{25+9}$
 $= \frac{34-34i}{34}$
 $= 1-i$

(ii) $|1-i| = \sqrt{1+(-1)^2} = \sqrt{2}$

(iii) $\text{Arg}(1-i) = -\frac{\pi}{4}$

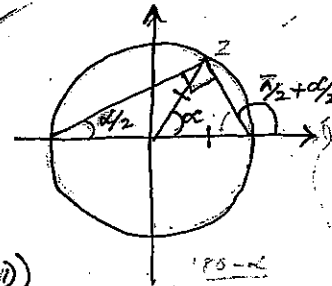


(d)



(i) $|Z^2 - Z| = |Z(Z-1)|$
 $= |Z| |Z-1|$
 $= |Z-1|$
 $\because |Z| = 1$ unit circle

(ii) $\text{Arg}(Z^2) = \text{Arg}(ZZ)$
 $= 2\text{Arg}(Z)$
 $= 2\alpha$



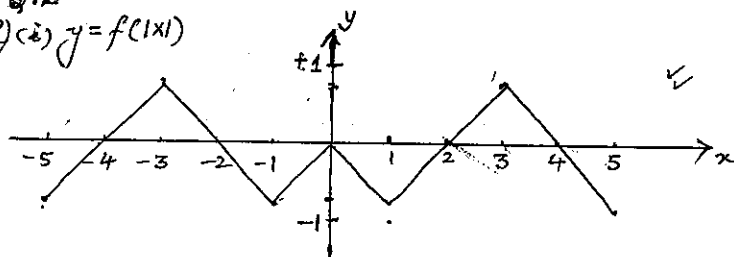
and

$\text{Arg}(Z^2 - Z) = \text{Arg}(Z(Z-1))$
 $= \text{Arg}(Z) + \text{Arg}(Z-1)$
 $= \alpha + \frac{\pi}{2} + \frac{\pi}{2}$
 $= \frac{3\alpha + \pi}{2}$

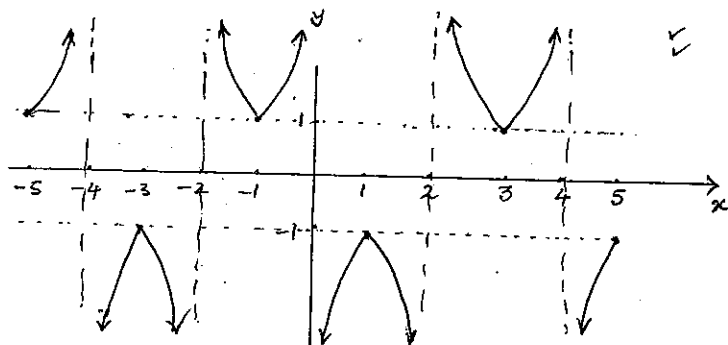
$\frac{180 - \alpha}{2}$
 $\frac{90 - \alpha}{2}$
 $180 - (90 - \frac{\alpha}{2})$
 $90 + \frac{\alpha}{2}$

Mark

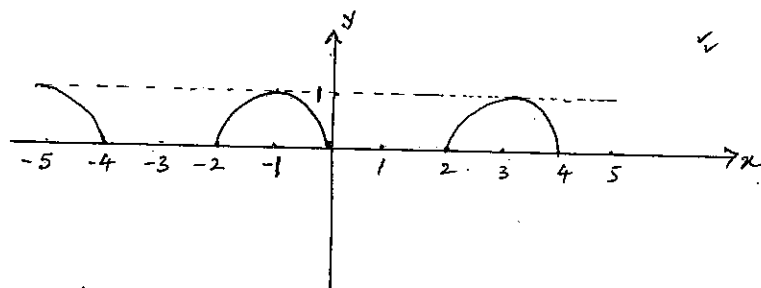
Q.2
(a) (i) $y = f(|x|)$



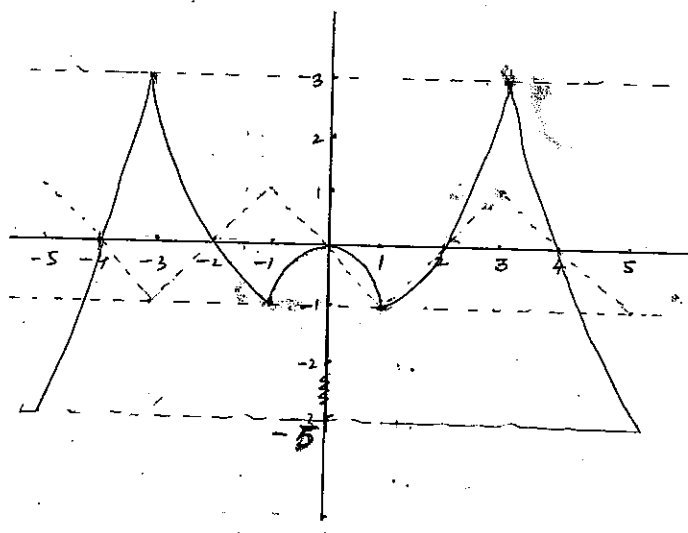
(ii) $y = \frac{1}{f(x)}$



(iii) $y = \sqrt{f(x)}$



(iv) $y = x f(x)$



(b) $y = \frac{9x-27}{x^2-x-2} = \frac{9(x-3)}{(x-2)(x+1)}$ $\therefore$ all real x
 $x \neq 2, -1$

x -intercept $x = 3$

$x = 0, y = 13/2$

Neither odd nor even

(ii) $y' = \frac{(x^2-x-2)(9) - (9x-27)(2x-1)}{(x^2-x-2)^2}$

$= \frac{9x^2 - 9x - 18 - 18x^2 + 54x + 9x - 27}{(x^2-x-2)^2}$

$= \frac{-9x^2 + 54x - 45}{(x^2-x-2)^2} = \frac{-9(x-5)(x-1)}{[(x-2)(x+1)]^2}$

Stationary pt. $y' = 0$

$-9(x^2 - 6x + 5) = 0$

$(x-5)(x-1) = 0$

$x = 5$ or 1

$x = 5, y = \frac{9(2)}{(3)(6)} = 1$

$x = 1, y = \frac{9(-2)}{(-1)(2)} = 9$

x	5^-	5	5^+
y'	> 0		< 0

$(5, 1)$ is a max pt.

x	1^-	1	1^+
y'	< 0		> 0

$(1, 9)$ is a min pt.

(i) Asymptotes

Vertical $x = 2, x = -1$

Horizontal $y = 0$

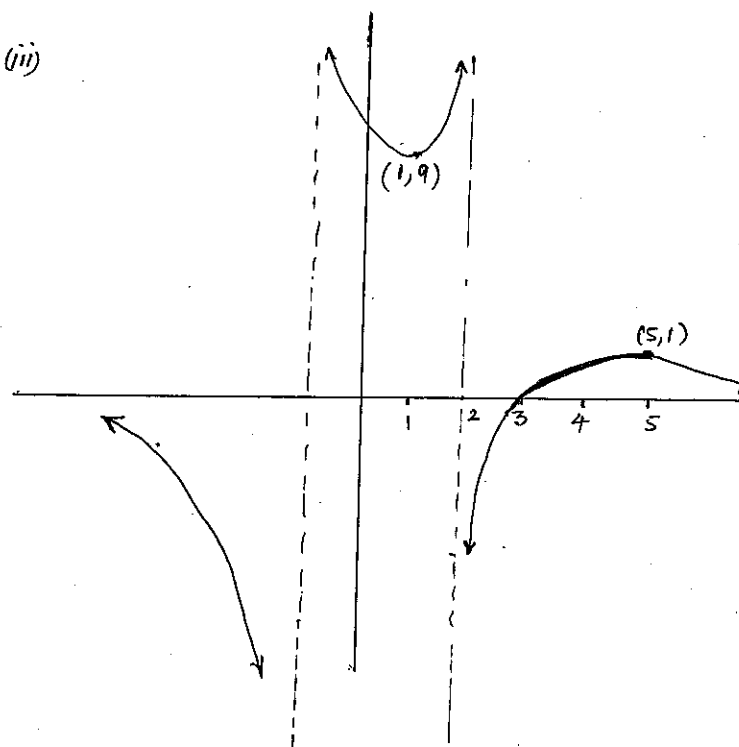
When $x = 2^+, y \rightarrow -\infty$

$x = 2^-, y \rightarrow \infty$

When $x \rightarrow -1^+, y \rightarrow \infty$

$x \rightarrow -1^-, y \rightarrow -\infty$

(iii)



Q3(c)

(i) $f(x) = \ln(\ln x)$

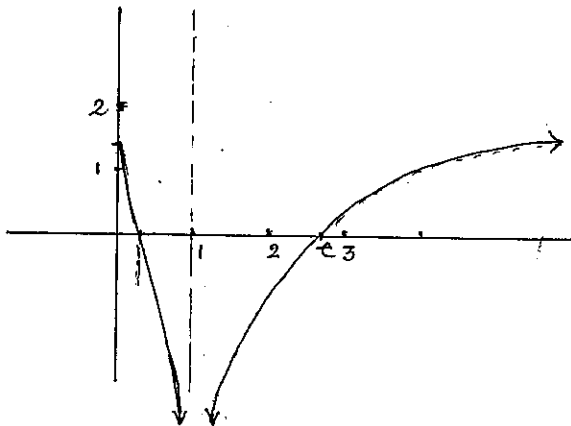
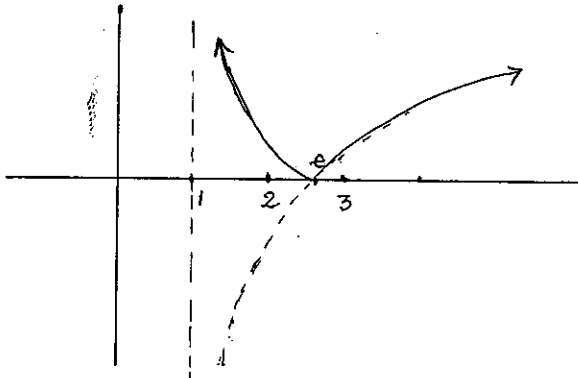
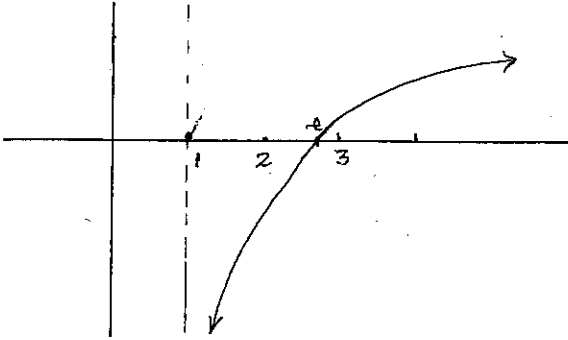
$\ln x > 0 \Rightarrow x > 1$ is the domain ✓

$$f'(x) = \frac{dy}{dx} = \frac{1}{\ln x} \cdot \frac{1}{x}$$

> 0 for all $x > 1$

$\therefore f(x)$ is increasing for all $x > 1$.

(ii)



Q3(a) $\frac{x^2}{9} + \frac{y^2}{16} = 1$

(i) $a=3, b=4$

$$e = \sqrt{1 - \frac{9}{16}} = \sqrt{\frac{7}{16}} = \frac{\sqrt{7}}{4}$$

Foci $(0, \pm be) \Rightarrow (0, \pm \sqrt{7})$ ✓

(ii) $y = \pm \frac{b}{a} \Rightarrow y = \pm \frac{4}{3}$

$$y = \pm \frac{16\sqrt{7}}{7}$$

(b) (i) $4x^2 + 9y^2 = 36$

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$8x + 18y \frac{dy}{dx} = -\frac{4x}{3y}$$

at $(1, \frac{4\sqrt{2}}{3})$ $\frac{dy}{dx} = \frac{-4 \times 1 \times 3}{9 \times 4\sqrt{2}}$

$$= -\frac{1}{3\sqrt{2}}$$

Equation of tangent is

$$y - \frac{4\sqrt{2}}{3} = -\frac{1}{3\sqrt{2}}(x-1)$$

$$3\sqrt{2}y - 4\sqrt{2} \times \frac{3\sqrt{2}}{3} = -x + 1$$

$$3\sqrt{2}y + x - 9 = 0$$

(ii) $3(\pm 6, 0)$ asymptotes $3y = \pm 4x$
 $\Rightarrow y = \pm \frac{4}{3}x$

$$ae = 6, \frac{b}{a} = \frac{4}{3}$$

$$b = \frac{4a}{3}$$

$$a^2 + b^2 = a^2 e^2$$

$$a^2 + \frac{16a^2}{9} = 36$$

$$9a^2 + 16a^2 = 9 \times 36$$

$$25a^2 = 9 \times 36$$

$$a^2 = \frac{9 \times 36}{25} = \frac{324}{25}$$

$$\therefore b^2 = \frac{16}{9} \times \frac{324}{25}$$

$$= \frac{576}{25}$$

$\therefore$ Equation is $\frac{x^2}{\frac{324}{25}} + \frac{y^2}{\frac{576}{25}} = 1$ ✓

$$400x^2 + 225y^2 = 5184$$

Q 3

$$(i) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, y = mx + c$$

$$(ii) (x) \quad bx^2 + a^2(m^2x^2 + 2mcx + c^2) = a^2b^2$$

$$(b^2 + a^2m^2)x^2 + 2a^2cmx + a^2(c^2 - b^2) = 0$$

$$\Delta = 0$$

$$\Rightarrow 4a^4c^2m^2 - 4a^2(c^2 - b^2)(b^2 + a^2m^2) = 0$$

$$\Rightarrow a^4c^2m^2 - a^2(c^2b^2 - b^4 + a^2m^2c^2 - a^2b^2m^2) = 0$$

$$\Rightarrow a^4c^2m^2 + a^2b^4 + a^4b^2m^2 - a^2b^2c^2 - a^4b^2m^2 = 0$$

$$\Rightarrow a^2b^2[b^2 + a^2m^2 - c^2] = 0$$

$$\Rightarrow c^2 = a^2m^2 + b^2$$

$$(A) \quad \frac{x^2}{16} + \frac{y^2}{9} = 1 \quad (3, 4)$$

$$y = mx + c$$

$$4 = 3m + c$$

$$c^2 = 16m^2 + 9$$

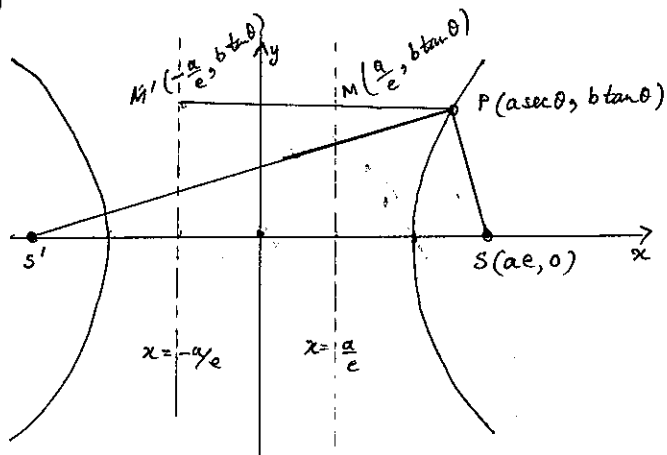
$$(4 - 3m)^2 = 16m^2 + 9$$

$$16 - 24m + 9m^2 = 16m^2 + 9$$

$$7m^2 - 24m + 7 = 0$$

$$\Rightarrow m_1, m_2 = \frac{c}{a} = -\frac{7}{7} = -1$$

(ii)



$$\frac{PS}{PM} = e \Rightarrow PS = ePM \quad \checkmark$$

$$\frac{PS'}{PM'} = e \Rightarrow PS' = ePM' \quad \checkmark$$

$$|PS - PS'| = e |PM - PM'| \quad \checkmark$$

$$= e \left| a \sec \theta - \frac{a}{e} - \frac{a}{e} - a \sec \theta \right|$$

$$= e \left| -\frac{2a}{e} \right|$$

$$= 2a \quad \checkmark$$

Q.3(d)

$$(i) \quad xy = c^2$$

$$y = \frac{c^2}{x}$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2} = -\frac{c^2}{c^2 q^2}$$

$$= -\frac{1}{q^2} \text{ at } Q(cq, \frac{c}{q}) \quad \checkmark$$

Normal at $Q(cq, \frac{c}{q})$ has slope q^2

Equation at Q

$$y - \frac{c}{q} = q^2(x - cq)$$

$$\text{i.e. } qy - c = q^3(x - cq) \quad \checkmark$$

(ii) The normal at Q intersects the hyperbola at $xy = c^2$ at P

$$\text{From (i)} \quad y = q^2x - cq^3 + \frac{c}{q}$$

at Point P

$$x(q^2x - cq^3 + \frac{c}{q}) = c^2$$

$$q^2x^2 - (cq^3 - \frac{c}{q})x - c^2 = 0$$

$$x^2 - \frac{c}{q^2}(cq^3 - \frac{1}{q})x - \frac{c^2}{q^2} = 0$$

$$x^2 - c(q - \frac{1}{q^3})x - \frac{c^2}{q^2} = 0 \quad \checkmark$$

The roots of this quadratic equation in x are the x -coordinates of Q and P .

(iii) Let the x -coordinate of P be X

then product of roots.

$$cq \cdot X = -\frac{c^2}{q^2}$$

$$= X = -\frac{c}{q^3}$$

substitute this in

$$xy = c^2$$

$$y = \frac{c^2}{-\frac{c}{q^3}} = -cq^3$$

The coordinates of P are. $\checkmark$

$$(-\frac{c}{q^3}, -cq^3)$$

(iv)

Equation of OQ is $y = mx$

$$m = \frac{\frac{c}{cq} - 0}{cq - 0} = \frac{1}{q^2}$$

The line $y = \frac{1}{q^2}x$ cuts $xy = c^2$ at P

$$\text{where } x \cdot \frac{1}{q^2}x = c^2$$

$$x^2 = q^2c^2$$

$$\therefore x = \pm cq$$

L has x -coordinate $-cq$

Sub. in $xy = c^2$

$$-cq \cdot y = c^2$$

$$y = -\frac{c}{cq}$$

i.e. L is $(-cq, -\frac{c}{cq})$ [can also be made by observation] $\checkmark$

$$m_{PL} = \frac{-\frac{c}{cq} + cq^3}{-cq + \frac{c}{q^3}}$$

$$= \frac{-\frac{1}{q} + q^3}{-q + \frac{1}{q^3}}$$

$$= \frac{-1 + q^4}{-q + \frac{1}{q^3}}$$

$$= \frac{-1 + q^4}{q} \times \frac{q^3}{-q^4 + 1}$$

$$= -q^2 \quad \checkmark$$

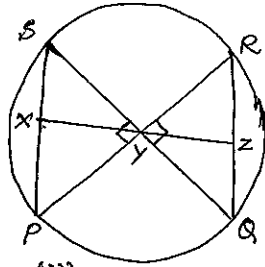
$$m_{QL} = \frac{\frac{c}{cq}}{cq} = \frac{1}{q^2}$$

$$m_{PL} \times m_{QL} = -q^2 \times \frac{1}{q^2}$$

$$= -1 \quad \checkmark$$

Therefore $\angle PLQ = 90^\circ$

Q4

(a) (i) $\angle SYP = 90^\circ$ (Given) $\therefore SP$ is the diameter of circle passing through Y . X is the mid pt. of SP (Given) $\therefore X$ is the centre $\therefore SX = XP = XY$ (equal radii)(ii) Let $\angle XSY = x$, $\angle XPY = y$ $\angle XSY = x$ (equal angles opp. to equal sides SX & XY of isosceles $\triangle SXY$)Similarly $\angle PYX = y$ $\therefore x + y = 90^\circ$ ($\because SYP = 90^\circ$ given) $\angle RYZ = y$ (vertically opp. to $\angle XYP$) $\angle YRZ = x$ (angles in the same segment standing on arc PQ)IN $\triangle RYZ$ $\angle RZY = 180 - (x + y)$ ($\angle$ sum of $\triangle$) $= 180 - 90^\circ$ ($\because x + y = 90^\circ$) $= 90^\circ$ $\therefore XZ \perp RQ$ (b) (i) $2x^3 - x^2 - 6x + 3 = 0$

$$\alpha + \beta + \gamma = \frac{1}{2}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = -3$$

$$\alpha\beta\gamma = -\frac{3}{2}$$

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$= (-3)^2 - 2(-\frac{3}{2})(\frac{1}{2})$$

$$= 9 + \frac{3}{2}$$

$$= \frac{21}{2}$$

(ii) $x^3 + 3x^2 + 7x + 9 = 0$ Let the roots be $\alpha, \alpha, \alpha + d$

$$\alpha + \alpha + \alpha + d = -3$$

$$3\alpha = -3$$

$$\therefore \alpha = -1$$

$$(-1-d)(-1) + (-1)(-1+d) + (-1-d)(-1+d) = 7$$

$$1 + d + 1 - d + 1 - d^2 = 7$$

$$1 - d^2 = 7 - 2$$

$$= 5$$

$$(1-d)(-1)(-1+d) = -9$$

$$- (1-d^2) = -9$$

$$-5 = -9$$

$$9 = 5 \quad \checkmark$$

(c) $2^{2^n} \geq 5^{2^n}$ for $n \geq 5$ Step 1. prove true for $n = 5$

$$\text{L.H.S.} = 2^{32}$$

TRUE

$$\text{R.H.S.} = 5^{10}$$

Step 2.

Assume true for $n = k$.

$$\text{i.e. } 2^{2^k} \geq 5^{2^k}$$

$$\text{or } 2^{2^k} - 5^{2^k} \geq 0$$

Step 3. Prove true for $n = k+1$

$$\text{i.e. } 2^{2^{k+1}} \geq 5^{2^{k+1}}$$

$$\text{i.e. } 2^{2^{k+1}} - 5^{2^{k+1}} \geq 0$$

$$\begin{aligned} \text{L.H.S. } 2^{2^k \cdot 2} &= (2^{2^k})^2 \\ &\geq (5^{2^k})^2 \\ &= 5^{4k} \\ &= 5^{2k+2k} \\ &\geq 5^{2k+10} \quad (\because k \geq 5) \\ &\geq 5^{2k+2} \\ &\neq 5^{2(k+1)} \end{aligned}$$

* Hence true for $n = k+1$.Hence by the principles of mathematical induction the statement is true $\forall n \geq 5$.